Article

Predicting Football Match Outcomes Using Machine Learning and Probability Calibration

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Abstract

Football match outcomes are influenced by a complex interplay of dynamic team strategies and stochastic match events, posing a significant challenge for predictive analytics. In this paper, we propose a unified benchmarking and calibration framework for football forecasting, utilizing a dataset of approximately 230,000 matches. We systematically compare diverse learning paradigms—ranging from linear models (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree, LightGBM, Random Forest) and Neural Network models (MLP). To enhance the reliability of our predictions, we implement the Newton–Raphson method to strip away bookmaker margins and incorporate an Isotonic Calibration layer for the subsequent derivation of robust betting signals. Our empirical results demonstrate that while methods like Random Forest excel in overall Precision (reaching 65.11% for Home Win and 69.07% for Away Win), the Multi-Layer Perceptron (MLP) yields superior non-linear feature interaction mapping, achieving the highest Recall across both markets (63.94% and 60.25% respectively). Our findings show that by combining advanced feature engineering with probability calibration, we can create more reliable models, helping to manage risk and identify betting value more effectively than standard statistical methods.

Keywords: Machine learning; neural networks; stochastic techniques.

1. Introduction

Many real world problems can be tackled by the so - called machine learning tools. Among these problems one can find cases derived from various scientific fields, such as physics [1,2], astronomy [3,4], chemistry [5,6], medicine [7,8], economics [9,10], image processing [11], time series forecasting [12] etc. Common machine learning tools are artificial neural networks [13,14], decision trees [15,16], support vector machines (SVM) [17,18] etc. In this research study, we address the problem of predicting the outcome of a football match by examining it through a variety of machine learning techniques. This problem was tackled by various researches during the past years.

Predicting football match outcomes is difficult because of the high levels of noise, changing team strategies, and unpredictable match events. Standard predictive models often struggle in these volatile environments. This research is important for three main

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reasons: First, it addresses the core issue of probability calibration in sports analytics. Machine learning models often produce raw scores that do not reflect real-world probabilities, especially when the data is noisy. By implementing methods like Isotonic Calibration and using the Newton–Raphson [19] approach to remove bookmaker margins, we provide a framework that transforms biased model outputs into reliable, objective probabilities. Second, we conduct a rigorous comparative analysis under strictly uniform conditions. Instead of focusing on just one algorithm, we test fundamentally different models—including linear models, GAM [20], tree structures (FastTree [21], LightGBM [22], Random Forest [23]), and Neural Network models (MLP)—using the exact same input features. This provides an unbiased look at how each model handles the non-linearity and volatility of sports data. Finally, from a practical standpoint, this research highlights the important trade-off between Precision and Recall in different betting markets. Whether we are analyzing Home Win and Away Win markets, finding the right balance is essential. Our results show that the Multi-Layer Perceptron (MLP), when integrated into our calibrated framework, consistently maximizes the recovery of hidden value (Recall) without sacrificing overall predictive accuracy. Ultimately, this work helps bridge the gap between simple statistical forecasting and more practical, value-driven risk management in sports betting.

For example, Hvattum et al. proposed the incorporation of ELO ratings for the prediction of match outcomes in their study [24]. Also, Owramipur et al. suggested the usage of Baysian networks [25] to predict the outcomes for the Spanish League Barcelona team [26]. Similarly, Presetio et al. proposed the incorporation of logistic regression [27] to predict the outcomes of football matches [28]. Moreover, Martins et al. used a polynomial classifier for the prediction of football outcomes [29]. Schauberger et al. selected the method of random forests to predict the results of football matches in their study [30]. Various attributes of football players was studied by Danisk et al. in order to predict the result of football matches in their related paper [31]. Constantinou proposed a new machine learning model, named Dolores which is based on Hybrid Bayesian Networks, to predict football match outcomes from datasets selected in a series of countries [32]. Baboota et al. applied gradient boosting [33] for English Premier League [34]. Additionally, a deep learning framework was designed from Rahman to tackle the problem of prediction football results [35].

To address the limitations of individual algorithms and the common issue of probability distortion in sports forecasting, we propose a unified benchmarking and calibration framework. Our methodology focuses on three core contributions:

- A Consistent Feature Engineering Pipeline: We build a predictive feature vector that combines dynamic team performance indicators—such as Elo ratings [24], Home Clean Sheet, and Away Fail-To-Score ratios—with raw bookmaker market odds. These odds are systematically adjusted using the Newton–Raphson method to isolate the intrinsic probabilities and remove market margins.
- Comprehensive Architectural Benchmarking: We conduct a comparative evaluation of different learning paradigms under strictly uniform conditions. By testing models ranging from linear approaches (Logistic Regression) and generalized additive frameworks (GAM) to tree structures (FastTree , LightGBM, Random Forest) and Neural Network models (MLP), we provide an unbiased look at how different models handle the high-dimensional volatility of football data.
- Framework for Robust Betting Signals: To bridge the gap between model scores and actionable betting decisions, our framework incorporates an Isotonic Calibration layer. As our next research phase, we apply this stabilization layer to transform raw output scores into mathematically objective probabilities, enabling the derivation of

robust betting signals. As empirical results demonstrate, this process highlights clear performance trends across both the Home Win and Away Win markets:

- Performance Stability: Linear models and methods, such as Logistic Regression and Random Forest provide high structural stability and competitive Precision (e.g., 65.11% for Home Win and 69.07% for Away Win).
- Non-Linear Interaction Mapping: The Multi-Layer Perceptron (MLP) acts as the most balanced model, consistently achieving the highest Recall in both markets (63.94% for Home Win and 60.25% for Away Win). This suggests that the MLP is better at mapping complex feature interactions and identifying "hidden value" opportunities that traditional statistical models often miss.

Ultimately, this work helps bridge the gap between simple statistical forecasting and more practical, value-driven risk management. By showing that the MLP maintains structural integrity while maximizing the recovery of betting value, we provide a reliable roadmap for building scalable and calibrated predictive systems in sports analytics.

The remaining of this article is organized as follows: in section 2 the proposed datasets and the used methodology are fully described, in section 3 the experimental results are outlined and finally in section 4 a discussion is provided based on the experimental results.

2. Materials and Methods

2.1. *The Dataset Employed*

In this study, we utilized a large-scale dataset of historical football matches sourced from a professional sports analytics repository. The data collection process prioritized high granularity, incorporating not only match results but also advanced performance indicators and pre-match market probabilities. The dataset's reliability is ensured through its internal consistency and widespread use in sports forecasting benchmarks.

2.2. *Dataset Description*

The experimental dataset comprises 225,474 match events spanning from 2018 to 2026. To ensure rigorous evaluation, the data was partitioned into a training set of 149,998 matches and a dedicated test set of 75,476 matches. The core variables included in the raw data are summarized in Table 1. The inclusion of *LeagueId* and *SeasonPhase* allows the model to account for competitive context and seasonal variations.

Table 1. Description of the original dataset features.

Feature	Description / Range
MatchDate	Temporal identifier (2018–2026)
League / LeagueId	Categorical identifiers for competitive tiers
HomeTeam / AwayTeam	Competing squad identifiers
Label	Primary outcome target (1, X, 2)
EloDiff	Difference in Elo ratings (-500 to $+500$)
ProbHome/Draw/Away	Pre-match market probabilities (0 to 1)
RankDifference	Difference in league table standings
SeasonPhase	Categorical marker for season stage
AttackVsDefense / DefenseVsAttack	Derived offensive and defensive interaction ratios
TotalScoringPotential	Normalized long-term scoring capacity
TotalDefensiveWeakness	Normalized long-term goal concession rate
HomeCleanSheetRatio	Probabilistic historical clean sheet frequency
AwayFTSRatio	Away team Failed-to-Score probability

2.3. *Feature Engineering and Mathematical Formulation*

To capture the dynamic interaction between offensive and defensive capacities, the raw data was transformed into formal performance metrics. Let $AvgH_s$ and $AvgH_c$ denote the average goals scored and conceded by the home team, and $AvgA_s$, $AvgA_c$ the corresponding values for the away team. A smoothing constant $\epsilon = 0.01$ was applied to prevent division by zero.

AttackVsDefense: Represents the home team's offensive efficiency relative to the away team's defensive record:

$$\text{AttackVsDefense} = \frac{AvgH_s}{AvgA_c + \epsilon} \tag{1}$$

DefenseVsAttack: Quantifies the home team's defensive capability against the away team's scoring force:

$$\text{DefenseVsAttack} = \frac{AvgH_c}{AvgA_s + \epsilon} \tag{2}$$

TotalScoringPotential: Modeled as the numerical difference between home and away scoring capacities, identifying the relative offensive edge:

$$\text{TotalScoringPotential} = AvgH_s - AvgA_s \tag{3}$$

TotalDefensiveWeakness: Assesses the relative vulnerability of the home defense compared to the away defense as a ratio:

$$\text{TotalDefensiveWeakness} = \frac{AvgH_c}{AvgA_c + \epsilon} \tag{4}$$

HomeCleanSheetRatio (HCSR) and ***AwayFTSRatio*** (AFTS) capture probabilistic historical frequencies:

$$HCSR = \frac{\sum_{i=1}^{N} \mathbb{I}(G_{H,c} = 0)}{N}, \quad AFTS = \frac{\sum_{j=1}^{N} \mathbb{I}(G_{A,s} = 0)}{N} \tag{5}$$

2.4. Feature Importance Motivation

The generated feature space is explicitly designed to satisfy three essential, complementary operational criteria:

1. Long-term structural team performance using rating systems (*EloDiff*, *RankDifference*).
2. Granular tactical interaction effects (*AttackVsDefense*, *DefenseVsAttack*).
3. Distribution-based prior probabilities derived from betting market evaluations (*ProbHome*, *ProbDraw*, *ProbAway*). This multi-source configuration mitigates model vulnerability to single-feature bias and improves generalization accuracy.

3. Results

3.1. Experimental settings

For the development and training of the predictive models for sports outcomes, the ML.NET [36] framework was employed—an open-source, cross-platform machine learning framework developed by Microsoft. The choice of ML.NET was driven by its capability to integrate sophisticated algorithms, such as FastForest (a highly optimized implementation of the Random Forest), which offers both computational efficiency and robustness when handling large-scale datasets. To evaluate the stability of the models, a 30-run stability analysis was conducted. This rigorous validation process ensured consistent performance metrics (Error, Precision, and Recall) and minimized the risk of overfitting, thereby validating the model's reliability for analyzing noisy, real-world data. The following methods were incorporated in the conducted experiments from ML.NET:

1. FastTree [21].
2. Gam [20].
3. LightGBM [22].
4. Logistic Regression [27].
5. MLP, that denotes an artificial neural network [13,14] with 10 processing nodes. This network is trained using the L-BFGS method [37].
6. Random Forest [23].
7. RBF, that denotes a Radial Basis Function network [38,39].

Table 2 presents the parameters used by the machine learning methods employed in the conducted experiments.

Table 2. Experimental settings.

METHOD	PARAMETER	VALUE
FASTTREE	Leaves	20
FASTTREE	Trees	600
FASTTREE	MinExample	100
FASTTREE	LearnRate	0.05
FASTTREE	FeatureFraction	0.9
RandomForest	Leaves	20
RandomForest	Leaves	100
RandomForest	MinExample	10
RandomForest	FeatureFraction	0.8
LightGBM	Leaves	31
LightGBM	MinExample	20
LightGBM	LearnRate	0.1
LightGBM	Iterations	100
GAM	Iterations	100
GAM	MaxBinCount	255
GAM	LearnRate	0.1
RBF	Clusters (K-means)	2
RBF	Sigma (Gaussian)	1.4
RBF	MaxIter	200
MLP	LBFGS Iterations	10
Logistic Regression	L2Regularization	0.1
Logistic Regression	Tolerance	10^{-4}

These parameters are analyzed in detail below:

1. **Leaves**: Defines the complexity of each tree by setting the maximum number of leaves. A higher value allows the model to capture more complex patterns but increases the risk of overfitting.
2. **Trees**: Specifies the total number of trees to build. Increasing this number generally improves model stability and predictive performance but requires more training time.
3. **MinExample**:Sets the minimum number of data samples required to create a leaf. This acts as a regularization mechanism; higher values prevent the model from learning from noise or outliers.
4. **LearnRate**: Controls the contribution of each tree to the final prediction. A lower learning rate makes the optimization process more robust, requiring more trees to reach an optimal solution.

5. **FeatureFraction**: Determines the percentage of features to consider when building each tree. Using a fraction (e.g., 0.8) introduces randomness, which helps the model avoid focusing too heavily on a single feature and improves generalization.
6. **MaximumIterations** (Iterations): The number of times the algorithm passes through the entire training dataset to update weights.
7. **MaxBinCount**: Sets the maximum number of bins used to group features, helping to discretize continuous variables for more efficient processing.
8. **Clusters** (K-Means): Defines the number of centroids (hidden neurons) used to represent the input space. Each cluster captures a local region of the data.
9. **Sigma** (Gaussian Spread): Controls the width of the radial basis function (Gaussian kernel). It determines how far the influence of a single neuron extends in the input space; a smaller sigma makes neurons more sensitive to localized inputs.
10. **L2Regularization**: A penalty applied to the magnitude of the model weights. It discourages overly large weights, which helps in preventing overfitting and improves the model's ability to generalize to unseen data.
11. **Tolerance**: The threshold for the optimization algorithm to determine convergence. It dictates how small the improvements in the loss function must be before the algorithm stops.

3.2. Experimental results

Table 3. Home win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	34.63%	64.91%	63.28%
GAM	34.89%	64.82%	62.75%
LightGBM	34.91%	64.59%	63.00%
Logistic Regression	34.51%	65.03%	63.44%
MLP	35.05%	64.21%	63.94%
Random Forest	34.71%	65.11%	62.86%
RBF	44.31%	54.75%	54.71%

Table 4. Away win experimental results.

MODEL	ERROR	PRECISION	RECALL
FastTree	27.97%	68.04%	59.29%
GAM	28.07%	68.64%	58.36%
LightGBM	27.96%	68.07%	59.31%
Logistic Regression	27.82%	68.21%	59.66%
MLP	27.90%	67.59%	60.25%
Random Forest	27.96%	69.07%	58.39%
RBF	32.62%	56.68%	52.91%

4. Conclusions

Author Contributions: M.S. and I.G.T. conducted the experiments, employing several datasets and provided the comparative experiments. M.S. and G.G. performed the statistical analysis and prepared the manuscript. All authors have read and agreed to the published version of the manuscript.

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References

1. Raissi, M., & Karniadakis, G. E. (2018). Hidden physics models: Machine learning of nonlinear partial differential equations. Journal of Computational Physics, 357, 125-141.
2. Kashinath, K., Mustafa, M., Albert, A., Wu, J. L., Jiang, C., Esmaeilzadeh, S., ... & Prabhat, N. (2021). Physics-informed machine learning: case studies for weather and climate modelling. Philosophical Transactions of the Royal Society A, 379(2194), 20200093.
3. Viquar, M., Basak, S., Dasgupta, A., Agrawal, S., & Saha, S. (2019). Machine learning in astronomy: A case study in quasar-star classification. Emerging Technologies in Data Mining and Information Security: Proceedings of IEMIS 2018, Volume 3, 827-836.
4. Luo, S., Leung, A. P., Hui, C. Y., & Li, K. L. (2020). An investigation on the factors affecting machine learning classifications in gamma-ray astronomy. Monthly Notices of the Royal Astronomical Society, 492(4), 5377-5390.
5. Meuwly, M. (2021). Machine learning for chemical reactions. Chemical Reviews, 121(16), 10218-10239.
6. J.A. Aguiar, M.L. Gong, T.Tasdizen, Crystallographic prediction from diffraction and chemistry data for higher throughput classification using machine learning, Computational Materials Science **173**, 109409, 2020.
7. S.S. Yadav, S.M. Jadhav, Deep convolutional neural network based medical image classification for disease diagnosis. J Big Data **6**, 113, 2019.
8. L. Qing, W. Linhong , D. Xuehai, A Novel Neural Network-Based Method for Medical Text Classification, Future Internet **11**, 255, 2019.
9. Athey, S. (2018). The impact of machine learning on economics. In The economics of artificial intelligence: An agenda (pp. 507-547). University of Chicago Press.
10. R. Hafezi, J. Shahrabi, E. Hadavandi, A bat-neural network multi-agent system (BNNMAS) for stock price prediction: Case study of DAX stock price, Applied Soft Computing **29**, pp. 196-210, 2015.
11. Ghai, D., Tripathi, S. L., Saxena, S., Chanda, M., & Alazab, M. (Eds.). (2022). Machine learning algorithms for signal and image processing. John Wiley & Sons.
12. Ahmed, N. K., Atiya, A. F., Gayar, N. E., & El-Shishiny, H. (2010). An empirical comparison of machine learning models for time series forecasting. Econometric reviews, 29(5-6), 594-621.
13. Zou, J., Han, Y., & So, S. S. (2009). Overview of artificial neural networks. Artificial neural networks: methods and applications, 14-22.
14. Abiodun, O. I., Jantan, A., Omolara, A. E., Dada, K. V., Mohamed, N. A., & Arshad, H. (2018). State-of-the-art in artificial neural network applications: A survey. Heliyon, 4(11).
15. De Ville, B. (2013). Decision trees. Wiley Interdisciplinary Reviews: Computational Statistics, 5(6), 448-455.
16. Kotsiantis, S. B. (2013). Decision trees: a recent overview. Artificial Intelligence Review, 39(4), 261-283.
17. Pisner, D. A., & Schnyer, D. M. (2020). Support vector machine. In Machine learning (pp. 101-121). Academic Press.
18. Valkenborg, D., Rousseau, A. J., Geubbelmans, M., & Burzykowski, T. (2023). Support vector machines. American journal of orthodontics and dentofacial orthopedics, 164(5), 754-757.
19. Clarke, S. (2016). A review of methods for deriving probability estimates from betting odds.
20. Wood, S. N. (2017). Generalized Additive Models: An Introduction with R (2nd ed.). Chapman and Hall/CRC.
21. Price, M. N., Dehal, P. S., & Arkin, A. P. (2010). FastTree 2–approximately maximum-likelihood trees for large alignments. PloS one, 5(3), e9490.
22. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. Advances in neural information processing systems, 30.
23. Liaw, A., & Wiener, M. (2002). Classification and regression by randomForest. R News, 2(3), 18-22.s
24. Hvattum, L. M., & Arntzen, H. (2010). Using ELO ratings for match result prediction in association football. International Journal of forecasting, 26(3), 460-470.
25. Marcot, B. G., & Penman, T. D. (2019). Advances in Bayesian network modelling: Integration of modelling technologies. Environmental modelling & software, 111, 386-393.

26. Owramipur, F., Eskandarian, P., & Mozneb, F. S. (2013). Football result prediction with Bayesian network in Spanish League-Barcelona team. International Journal of Computer Theory and Engineering, 5(5), 812.
27. Stoltzfus, J. C. (2011). Logistic regression: a brief primer. Academic emergency medicine, 18(10), 1099-1104.
28. D. Prasetio and D. Harlili, "Predicting football match results with logistic regression," 2016 International Conference On Advanced Informatics: Concepts, Theory And Application (ICAICTA), Penang, Malaysia, 2016, pp. 1-5, doi: 10.1109/ICAICTA.2016.7803111.
29. Martins, R. G., Martins, A. S., Neves, L. A., Lima, L. V., Flores, E. L., & do Nascimento, M. Z. (2017). Exploring polynomial classifier to predict match results in football championships. Expert Systems with Applications, 83, 79-93.
30. Schauberger, G., & Groll, A. (2018). Predicting matches in international football tournaments with random forests. Statistical Modelling, 18(5-6), 460-482.
31. Danisik, N., Lacko, P., & Farkas, M. (2018, August). Football match prediction using players attributes. In 2018 World symposium on digital intelligence for systems and machines (DISA) (pp. 201-206). IEEE.
32. Constantinou, A. C. (2019). Dolores: a model that predicts football match outcomes from all over the world. Machine learning, 108(1), 49-75.
33. Natekin, A., & Knoll, A. (2013). Gradient boosting machines, a tutorial. Frontiers in neurorobotics, 7, 63623.
34. Baboota, R., & Kaur, H. (2019). Predictive analysis and modelling football results using machine learning approach for English Premier League. International Journal of Forecasting, 35(2), 741-755.
35. Rahman, M. A. (2020). A deep learning framework for football match prediction. SN Applied Sciences, 2(2), 165.
36. Microsoft. (2026). ML.NET: Machine Learning for .NET. Retrieved June 30, 2026, from https://dotnet.microsoft.com/en-us/apps/machinelearning-ai/ml-dotnet
37. Liu, D. C., & Nocedal, J. (1989). On the limited memory BFGS method for large scale optimization. Mathematical programming, 45(1), 503-528.
38. J. Park and I. W. Sandberg, Universal Approximation Using Radial-Basis-Function Networks, Neural Computation **3**, pp. 246-257, 1991.
39. G.A. Montazer, D. Giveki, M. Karami, H. Rastegar, Radial basis function neural networks: A review. Comput. Rev. J **1**, pp. 52-74, 2018.